ALESSANDRO TANSINI

Researcher, Data Science & Policy Support | Transport Decarbonisation

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Date and place of birth: 06/08/1990, Milan, Italy

Links: ORCID, Google scholar, Researchgate, Scopus, LinkedIn

PROFESSIONAL SUMMARY

Energy engineer with a PhD in Energetics and over ten years of experience in scientific research on road vehicles CO_2 emissions, electrification, innovative vehicle technologies and sustainable transport. Deep knowledge of data collection and analysis, including: experimental measurements, OBD/CAN, GNSS, telematics, web scrapping, data reporting. Currently a contract agent at the European Commission's Joint Research Centre (JRC), specialised in real-world monitoring, modelling, and policy support for vehicle fuel/electricity consumption and energy efficiency. Author of peer-reviewed publications in leading energy and environmental journals and contributor to vehicle regulations.

PROFESSIONAL EXPERIENCE

Sep 2020 – present	**Scientific Policy Officer – European Commission's Joint Research Centre (JRC), Ispra, Italy** *Sustainable, Smart and Safe Transport Unit | Real-world emissions monitoring and modelling.* • Review, collection and analysis of literature on road vehicle CO_2 emissions and energy consumption • Design and execution of experimental campaigns for battery electric, plug-in hybrid electric, hydrogen fuel cell vehicles and others • Design of an experimental campaign based on telemetry data from IoT devices with automatic data ingestion, correction, processing and reporting • Development and maintenance of CO_2 calculation tools extended to innovative powertrains (BEV, PHEV, H_2) • Analysis of monitoring data under EU Regulation 2021/392; reporting on real-world fuel consumption (340201/2019/817718/SER/CLIMA.C.4) • Scientific evaluation of amendments to EU Regulation 2017/1151 regarding PHEVs Utility Factors, CO_2 emissions and energy consumption • Assessment of Vehicle Integrated PhotoVoltaics (VIPV) environmental impact • Amendment of UN R154 with respect to the definition of energy into the vehicle for externally chargeable vehicles (BEV, PHEV) for On-Board Fuel and energy Consumption Monitoring (OBFCM) • Supported discussions at European and United Nation levels for the modelling of in-use consumption from PHEVs for Life Cycle Assessment regulations • Analysis of M1 and N1 monitoring data from vehicles registered in Europe for the development of simple CO_2 emissions calculation formula in support of the European Individual Vehicle Approval (IVA) scheme

Dec 2019 – Sep 2020	**External Expert – European Commission's Joint Research Centre (JRC),**

	Ispra, Italy *Ad-hoc expertise on hybrid vehicle measurement, operation, and data collection & analysis* • Literature review on hybrid electric vehicle CO_2 emissions and energy consumption • Dedicated experimental campaigns (vehicle testing in lab and on-road) • CO_2 prediction tools at vehicle and fleet level; monitoring exercise under EU Reg. 2019/631

Dec 2016 – Dec 2019	**Research Fellow (Grantholder 20) – European Commission's Joint Research Centre (JRC), Ispra, Italy** *Development of flexible calculation tools for the road-vehicles CO_2 EU regulatory framework* • Literature review on CO_2 emissions calculation and monitoring for road vehicles • Design and execution of experimental vehicle testing campaigns (lab and on-road) • Tools for CO_2 emissions of specific vehicle configurations and innovative technologies • Pre-pilot air drag procedure testing for EU Reg. 2017/2400 (heavy-duty vehicles) • Development of HDV fleet CO_2 emissions baseline for emission targets • Update of EU Reg. 2017/1152 and 1153 to include hybrid electric vehicles

Jul 2016 – Dec 2016	**Junior Calibration and Emissions Engineer – Positech Srl (now ALTEN), at Maserati Spa, Milan, Italy** *Vehicle Calibration Department – automotive engineering consultancy* • Development and refinement of control strategies for thermal management and CO_2 emissions on hybrid vehicles • NEDC (ECE R.83 and R.101) compliance testing; vehicle instrumentation, laboratory and on-road tests • Calibration and compliance checks; scientific analyses and reporting

Nov 2015 – May 2016	**Emissions Engineer Trainee – MV Agusta Motor Spa, Varese, Italy** *R&D Department – automotive manufacturing (motorbikes)* • CO_2 and pollutant measurements under WMTC for regulatory compliance • Use and maintenance of emission test equipment (roller test bench, gas analysers, computer lab automation) • Test report creation; participation in CUNA Circuit Emission Laboratories Correlation activities

2008 – 2015	**Private Tutor – Scientific subjects (high school and university level)**

Jun–Jul 2008	**Internship – Gamma Servizi Srl: customisation and installation of computer stations**

Jun 2007	**Internship – Contrel Srl, Lodi: quality control of electronic devices**

EDUCATION

Nov 2016 – Mar 2020	**PhD in Energetics – Politecnico di Torino (DENERG), Italy** *Thesis: Development of flexible calculation approaches to support the European CO_2 emissions regulatory scheme for road vehicles* Awarded: Dottorato di Ricerca (DR) – ISCED/EQF level 8

Oct 2012 – Jul 2015	**MSc in Power Production (Energy Engineering) – Politecnico di Milano, Italy** Focus: fuel properties, combustion processes, internal combustion engines, multi-physics fluid dynamics; CFD *Thesis: Low-temperature and high-efficiency combustion CFD modelling in a diesel engine for industrial applications* Final mark: 105/110 | ITA Diploma di Laurea Specialistica – ISCED/EQF level 7

Jul 2017	**Stuttgart International Summer School – FKFS, Stuttgart, Germany** System Integration, Simulation and Energy Management of Hybrid Electric Vehicles

Nov 2014	**ATHENS Programme – ENSTA ParisTech, Paris, France** Introduction to Vehicle Dynamics

Feb 2013 – Jul 2013	**ERASMUS Exchange – Universidad de Las Palmas de Gran Canaria, Spain**

Sep 2009 – Sep 2012	**BSc in Energy Engineering – Politecnico di Milano, Italy** Final mark: 94/110 | ITA Diploma di Laurea – ISCED/EQF level 6

Sep 2004 – Jul 2009	**IT Technician Diploma – IIS A. Volta, Lodi, Italy** 5-year programme: programming, telecommunications, IT systems & networks, electronics, mathematics Final mark: 91/100 | ISCED level 3

PUBLICATIONS

Peer-reviewed articles and technical reports (reverse chronological order).

[2026] G. Albano, K. Mattas, D. Komnos, A. Tansini, S. Vass, R. Donà, A. Garus, G. Fontaras, B. Nahmias-Biran, B. Ciuffo, **"Unveiling the energy implications of automated driving: Evidence from real-world data"** , *Journal of Cleaner Production, Volume 565, 2026,*
DOI: https://doi.org/10.1016/j.jclepro.2026.148294

[2026] M. Lopez-Juarez, A. Tansini, G. Di Pierro, R. Novella, G. Fontaras, **"Influence of the driving pattern and hydrogen mix scenario of next-generation FCV on performance and environmental impact in the 2025–2050 timeframe"** , *Energy Conversion and Management: X, Volume 30, 2026,*
DOI: https://doi.org/10.1016/j.ecmx.2026.101876

[2026] Pavlovic, J., Komnos, D., Suarez Corujo, J., Ktistakis, M., Passier, G., Tansini, A. et al. **"Risk assessment for the 2026 In-Service Verification (ISV) of CO2 emissions of Light-Duty Vehicles"** , *Publications Office of the European Union, 2026,*
DOI: https://data.europa.eu/doi/10.2760/2376420

[2025] Suarez Corujo J., Tansini A., Ktistakis M., Laverde Marin A., Komnos D., Pavlovic J., Fontaras G. **"Towards zero CO_2 emissions: Insights from EU vehicle on-board data."** *Science of the Total Environment.*
DOI: 10.1016/j.scitotenv.2025.180454

[2025] Tansini A., Laverde Marin A., Suarez Corujo J., Aguirre N., Fontaras G. **"Learning from the real-world: Insights on light-vehicle efficiency and CO_2 emissions from long-term on-board fuel and energy consumption data collection."** *Energy Conversion and Management.*
DOI: 10.1016/j.enconman.2025.119816

[2025] Mourtzouchou, A., Marin, A.L., Laveneziana, L., Tansini, A. et al. **"Comparative analysis of public and expert perceptions of electrified vehicles in the European Union"**, *Sci Rep 15, 21695 (2025).*
DOI: https://doi.org/10.1038/s41598-025-06071-0

[2025] Komnos, D., Tansini, A., Trentadue, G., Fontaras, G., Grigoratos, T., & Giechaskiel, B. (2025). **"Friction and Regenerative Braking Shares Under Various Laboratory and On-Road Driving Conditions of a Plug-In Hybrid Passenger Car"**. Energies, 18(15), 4104.
DOI: https://doi.org/10.3390/en18154104

[2025] Marin, A. L., Martínez-Plumed, F., Makridis, M. A., Tansini, A., Pulvirenti, L., Suarez Corujo, J., Fontaras, G. **"Follow the leader: a deep reinforcement learning framework for safe and efficient autonomous car-following"**. Journal of Intelligent Transportation Systems, 1–22.
DOI: https://doi.org/10.1080/15472450.2025.2576907

[2025] Tansini, A., Sartori Vieira, F., Ellis, H., Dunlop, E., Fontaras, G., **"Benchmarking the real-world energy benefits of vehicle-integrated photovoltaics"**, Conference : Transport and Pollution International Conference 2025, Paris, France. Conference proceedings under publication.

[2025] A. Bouter, J. Suarez, L. Laveneziana, D. Komnos, A. Tansini et al., **"A Comprehensive Analysis of Energy Use and GHG Emissions in EU Light-Duty Vehicles"**, Conference : Transport and Pollution International Conference 2025, Paris, France. Conference proceedings under publication.

[2025] **"Risk assessment for the 2025 in-service verification (ISV) of CO2 emissions of light-duty vehicles – Risk assessment ISV 2025 according to Commission Implementing Regulation (EU) 2023/2866"**, *Publications Office of the European Union,*
DOI: https://data.europa.eu/doi/10.2760/0897088

[2025] J. Suarez, M.A. Ktistakis, D. Komnos, A. Tansini, A.L. Marin, M. Makridis, B. Ciuffo, G. Fontaras, **"The impact of drivers' acceleration style on the vehicle energy performance: a real-world case study",** Transportation Research Procedia, Volume 82, 2025, Pages 3300-3319, ISSN 2352-1465,
DOI: https://doi.org/10.1016/j.trpro.2024.12.093

[2024] **"Risk assessment for the 2024 in-service verification (ISV) of CO2 emissions of light-duty vehicles – Risk assessment ISV 2024 according to Commission implementing regulation (EU) 2023/2866"**, Publications Office of the European Union, 2024
DOI: https://data.europa.eu/doi/10.2760/0897088

[2024] Pavlovic J., Tansini A., Suarez Corujo J., Fontaras G. **"Influence of vehicle and battery ageing and driving modes on emissions and efficiency in Plug-in hybrid vehicles",** *Energy Conversion and Management: X.*
DOI: https://data.europa.eu/doi/10.2760/574267

[2024] Di Pierro G., Bitsanis E., Tansini A., Bonato C., Martini G., Fontaras G. **"Fuel Cell Electric Vehicle Characterisation under Laboratory and In-use Operation",** *Energy Reports.*
DOI: 10.1016/j.egyr.2023.12.013

[2024] Gil-Sayas S., Di Pierro G., Tansini A., Serra S., Currò D., Broatch A., Fontaras G. **"Energy consumption of mobile air-conditioning systems in electrified vehicles under different ambient temperatures"** *International Journal of Engine Research.*
DOI: 10.1177/14680874231171303

[2023] Tansini A., Fontaras G., Millo F. **"A Multipurpose Simulation Approach for Hybrid Electric Vehicles to Support the European CO_2 Emissions Framework",** *Atmosphere, Vol. 14, Art. 587.*
DOI: [10.3390/atmos14030587](https://doi.org/10.3390/atmos14030587)

[2023] Bitsanis E., Broekaert S., Tansini A., Savvidis D., Fontaras G. **"Experimental evaluation of VECTO hybrid electric truck simulations",** *SAE Technical Papers.*
DOI: [10.4271/2023-01-0485](https://doi.org/10.4271/2023-01-0485)

[2023] Tansini, A., Suarez, J., Aguirre, N., Laverde, A., Fontaras, G., **"From physical testing to on-board fuel consumption monitoring and telemetry: a pilot project for capturing the real-world fuel consumption of vehicles" ,** *Proceedings of the 25th international transport & air pollution (TAP) and the 3rd shipping & environment (S&E) conferences, Publications Office of the European Union, 2024,* *https://data.europa.eu/doi/10.2760/564701*

[2023] Suarez, J., Laverde, A., Tansini, A., Ktistakis, M.A., Komnos, D., Fontaras, G., **"Observations on the Driving of Plug-In Hybrid Cars in Real-World Conditions"**. *In: Nathanail, E.G., Gavanas, N., Adamos, G. (eds) Smart Energy for Smart Transport. CSUM 2022. Lecture Notes in Intelligent Transportation and Infrastructure. Springer, Cham.*
DOI: https://doi.org/10.1007/978-3-031-23721-8_12

[2022] Tansini A., Pavlovic J., Fontaras G. **"Quantifying the real-world CO_2 emissions and energy consumption of modern plug-in hybrid vehicles"** *Journal of Cleaner Production, Vol. 362.*
DOI: [10.1016/j.jclepro.2022.132191](https://doi.org/10.1016/j.jclepro.2022.132191)

[2022] Thiel C., Gracia Amillo A., Tansini A., Tsakalidis A., Fontaras G., Dunlop E., Taylor N., Jäger-Waldau A., Araki K., Nishioka K., Ota Y., Yamaguchi M. **"Impact of climatic conditions on prospects for integrated photovoltaics in electric vehicles"** *Renewable and Sustainable Energy Reviews, Vol. 158.*
DOI: [10.1016/j.rser.2022.112109](https://doi.org/10.1016/j.rser.2022.112109)

[2022] Ktistakis M.A., Tansini A., Marin A.L., Suarez J., Komnos D., Fontaras G., **"Understanding the fuel consumption of plug-in hybrid electric vehicles: a real-world case study",** *Conference: THIESEL 2022. Conference on Thermo-and Fluid Dynamics of Clean Propulsion Powerplants*
DOI: [10.4995/Thiesel.2022.632801](https://doi.org/10.4995/Thiesel.2022.632801)

[2022] Tansini, A., Di Pierro, G., Fontaras, G., Gil-Sayas, S., et al., **"Battery Electric Vehicles Energy Consumption Breakdown from On-Road Trips 1",** *SAE Int. J. Adv. & Curr. Prac. in Mobility 5(3):977-987, 2023,*
DOI: https://doi.org/10.4271/2022-37-0009

[2022] Di Pierro, G., Tansini, A., Fontaras, G., and Bonato, C., **"Experimental Assessment of Powertrain Components and Energy Flow Analysis of a Fuel Cell Electric Vehicle (FCEV)",** *CO2 Reduction for Transportation Systems Conference, Turin, Italy, June 21, 2022,*
DOI: https://doi.org/10.4271/2022-37-0011

[2021] Zacharof N., Fontaras G., Ciuffo B., Tansini A., Prado-Rujas I. **"An estimation of heavy-duty vehicle fleet CO_2 emissions based on sampled data."** *Transportation Research Part D: Transport and Environment, Vol. 94.*
DOI: [10.1016/j.trd.2021.102784](https://doi.org/10.1016/j.trd.2021.102784)

[2020] Tansini A., Fontaras G., Millo F. **"A Theoretical and Experimental Analysis of the Coulomb Counting Method and of the Estimation of the Electrified-Vehicles Electricity Balance in the WLTP."** *SAE Technical Paper 2020-37-0020.*
DOI: [10.4271/2020-37-0020](https://doi.org/10.4271/2020-37-0020)

[2019] Suarez-Bertoa R., Pavlovic J., Trentadue G., Otura-Garcia M., Tansini A., Ciuffo B., Astorga C. **"Effect of Low Ambient Temperature on Emissions and Electric Range of Plug-In Hybrid Electric Vehicles."** *ACS Omega.*
DOI: [10.1021/acsomega.8b02459](https://doi.org/10.1021/acsomega.8b02459)

[2019] Di Pierro G., Millo F., Tansini A., Fontaras G., Scassa M. **"An Integrated Experimental and Numerical Methodology for Plug-In Hybrid Electric Vehicle 0D Modelling."** *SAE Technical Paper 2019-24-0072.*
DOI: [10.4271/2019-24-0072](https://doi.org/10.4271/2019-24-0072)

[2019] Zacharof N., Tansini A., Prado Rujas I., Grigoratos T., Fontaras G. **"A Generalized Component Efficiency and Input-Data Generation Model for Creating Fleet-Representative Vehicle Simulation Cases in VECTO."**

SAE Technical Paper 2019-01-1280.
DOI: [10.4271/2019-01-1280](https://doi.org/10.4271/2019-01-1280)

[2019] Doulgeris, S., Tansini, A. et al. **"Simulation-based assessment of the CO_2 emissions reduction potential from the implementation of mild-hybrid architectures on passenger cars to support the development of CO2MPAS."**

[2019] Tansini A., Fontaras G., Ciuffo B., Millo F., Prado Rujas I., Zacharof N. **"Calculating Heavy-Duty Truck Energy and Fuel Consumption Using Correlation Formulas Derived From VECTO Simulations."** *SAE Technical Paper 2019-01-1278.*
DOI: [10.4271/2019-01-1278](https://doi.org/10.4271/2019-01-1278)

[2018] Tansini A., Zacharof N.-G., Prado Rujas I., Fontaras G. **"Analysis of VECTO data for Heavy-Duty Vehicles (HDV) CO_2 emission targets."** *JRC Scientific and Technical Report, Publications Office of the European Union.*
DOI: [10.2760/551250](https://doi.org/10.2760/551250)

[2017] Zacharof N., Fontaras G., Ciuffo B., Tansini A., Grigoratos T., Prado Rujas I., Anagnostopoulos K. **"CO_2 emissions of the European Heavy Duty Truck Fleet: a Preliminary Analysis of the Expected Performance Using VECTO Simulator and Global Sensitivity Analysis Techniques."** *22nd International Transport and Air Pollution Conference, Zürich, Switzerland.*

[2017] Pavlovic J., Tansini A., Fontaras G., Ciuffo B., Otura Garcia M., Trentadue G., Suarez Bertoa R., Millo F. **"The Impact of WLTP on the official fuel consumption and electric range of Plug-in Hybrid Electric Vehicles in Europe."** *SAE International Journal of Passenger Cars – Electronic and Electrical Systems.* SAE Technical Paper 2017-24-0133

AWARDS

2019 - PhD prize - Doctoral School, Politecnico di Torino

Best 3-Minute Research Video (32nd Doctoral Cycle). Recognized for excellence in distilling complex research into an engaging and accessible format.

2025 – JRC Award for Excellence 2025 in the category "Best Team Collaboration"

I supported the organization of the 5th European Conference on Connected and Automated Driving (EUCAD 2025) that took place at the European Commission's Joint Research Centre in Ispra in May 2025. My contribution consisted in:

- *the management of the specialized thematic zone regarding Vehicle-Integrated PhotoVoltaics and*
- *presenting practical applications of fuel consumption monitoring through telematics to executive visitors.*

PUBLISHED RESOURCES

Tools and collections of data to which I contributed.

Vehicle Data Processing (VDP) tool (main developer, 2022-2026)

https://code.europa.eu/jrc-ldv/vdp
Flexible tool for the processing of experimental, naturalistic or simulated data regarding road vehicles usage.
It includes algorithms for data cleaning, GNSS/CAN data processing, powertrain calculations and more (e.g. Mobile Air Conditioning, braking energy regeneration and usage of friction brakes). It includes the Cycle Energy Demand (CED) tool supporting Implementing Regulation 2023/2866/EU on In-Service Verification of CO_2 emissions in Europe.

LEGENT data repository (main contributor, 2025-2026)

https://code.europa.eu/jrc-legent/legent-data

A living data repository of transport-and-driving-related data, including:

- *aggregated trips data from an automated data ingestion pipeline of OBD/OBFCM and telematics data (Tansini et al. 2025, 10.1016/j.enconman.2025.119816)*
- *driving data from three vehicles (Marin et al. 2025, https://doi.org/10.1080/15472450.2025.2576907)*
- *fuel consumption and driving intensity from OBFCM data of the European M1 vehicles fleet (Suarez et al. 2025, 10.1016/j.scitotenv.2025.180454*

CONFERENCES and EVENTS

Participation in conferences, live events, seminars and workshops with an active contribution.

- *CO2 Reduction for Transportation Systems Conference, 9-10 June 2026, Turin, Italy*
- *Transport Research Arena, 18-21 May 2026, Budapest, Hungary*
- *Transport and Pollution Conference, 4-6 November 2025, Rueil-Malmaison, France*
- *SAE CO_2 Reduction for Transportation Systems Conference, 12-13 June 2024, Turin, Italy*
- *Seminar on Vehicle-Integrated PhotoVoltaics and Road Show event, Joint Research Centre, 12 July 2024, Ispra, Italy*
- *Transport and Air Pollution Conference, Gothenburg, Sweden, 2023*
- *European Researchers Night, 29 September 2023, Museo della Scienza e della Tecnica, Milan, Italy (presenting on-board fuel consumption monitoring through telematics)*
- *THIESEL Conference, 13-16 September 2022, Valencia, Spain*
- *SAE CO_2 Reduction for Transportation Systems Conference, 21-22 June 2022, Turin, Italy*
- *"Science is everywhere" seminars for high school students, digital online seminars, 3 events in 2022*
- *SAE CO_2 Reduction for Transportation Systems Conference, 7-9 July 2020, Turin, Italy*
- *Focus Live Festival, 21-24 November 2019, Milan, Italy (presenting "Green Driving" in a thematic zone)*
- *Focus Live Festival, 1-2 June 2019, Genoa, Italy (presenting "Green Driving" in a thematic zone)*
- *WCX 2019 World Congress Experience, 9-11 April 2019, Detroit, USA*
- *SAENA Workshop, 22-23 November 2018, Reggio Emilia, Italy (presenting PHEV regulation and implications)*
- *PTNSS Congress 2017, 27-29 June 2017, Poznan, Poland*
- *SAE ICE Conference, 10-14 September 2017, Capri, Italy*

LANGUAGES

Italian: C2, **English:** C1 (TOEFL iBT 102/120, 2016), **Spanish:** B2 (Inter-institutional language course)

TECHNICAL SKILLS

Programming languages: Python, Matlab, Arduino, C and C++ (basic knowledge)

Simulation & measurements: Self-developed tools, Matlab/Simulink/Stateflow, ETAS (INCA, MDA, INTECRIO), ATI Vision, AVL IndiCom, Dewesoft, Kvaser, CSS electronics, Freematics, OBD, CAN

Data, Tools & Skills: data pipelines, data scrapping, data visualization, MongoDB, Git, Claude Code, IDEs (PyCharm, Visual Studio Code, Jupyter Lab, Jupyter Notebooks), CO2MPAS (EU Reg. 2017/1152–1153), VECTO (EU Reg. 2017/2400)

OS: Windows, Linux

Certificates: European Computer Driving Licence (ECDL)

OTHER INFORMATION

Driving Licence: type A and B; car and motorbike track driving experience

Volunteering: youth association, organisation of local events (2004–2009), plogging

Interests: motor sports, outdoor activities, photography, music, cooking